Exam 3

1. A hydrocarbon contains only the elements

a) carbon and oxygen
b) carbon and hydrogen
c) carbon, hydrogen, and oxygen
d) hydrogen and nitrogen

2. Which of the following represents a saturated hydrocarbon?

a) $CH_2=CH–CH_3$
b) $CH_3–CH_2–CH_3$
c) $CH{\equiv}CH$
d) C_6H_6

3. Which structural formula corresponds to butane?

a) $CH_3–CH=CH–CH_3$
b) $CH_3–CH_2–CH_2–CH_3$
c) $CH_3–C{\equiv}CH$
d) $CH_3–CH_2–CH_2–OH$

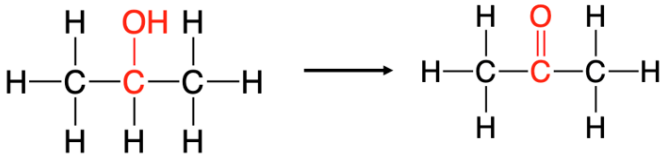
4. According to IUPAC rules, the main chain is

a) the shortest chain
b) the chain with the most substituents
c) the longest continuous chain regardless of bends
d) the most branched chain

5. Which one of the following compounds can show cis–trans isomerism?

a) CH_4
b) $CH_3CH=CHCH_3$
c) C_6H_6
d) $CH_3C{\equiv}CH$

6. What type of reaction is shown below?



a) Combustion
b) Oxidation
c) Halogenation
d) Dehydration

7. The structural formula of benzene is best represented as

a) ring of six carbons with alternating double bonds
b) six carbon ring with a circle in the center
c) ring of five carbons with one double bond
d) hexane

8. The functional group of an alcohol is

a) –COOH
b) –SH
c) –OH
d) –CO–

9. A secondary (2°) alcohol has the –OH group attached to a carbon

a) bonded to one alkyl group
b) bonded to two alkyl groups
c) bonded to three alkyl groups
d) in a ring only

10. Which of the following represents the correct result from the oxidation of a primary alcohol?

a) Alkene
b) Aldehyde
c) Ketone
d) Carboxylic acid

11. Thiols contain the functional group

a) –SH
b) –OH
c) –CHO
d) $–NH_2$

12. Which of the following describes the general structure of an ether?

a) R–OH
b) R–COOH
c) R–O–R′
d) $R–NH_2$

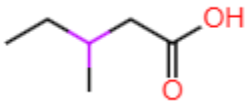
13. Which compound contains a carbonyl group at the end of the chain?

a) ketone
b) aldehyde
c) ether
d) carboxylic acid

14. The reduction of a ketone produces

a) a primary alcohol
b) an aldehyde
c) a secondary alcohol
d) a tertiary alcohol

15. Provide the correct IUPAC name for the following compound:

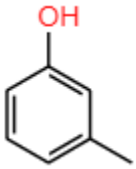


a) butanoic acid
b) 2-methylpropanoic acid
c) 3-methylpentanoic acid
d) pentanoic acid

16. Carboxylic acids are classified as acids according to which definition?

a) Lewis
b) Arrhenius
c) Brønsted–Lowry
d) None of the above

17. What is the name of the following compound?



a) 3-methylphenol
b) 3-hydroxyltoluene
c) 3-methyl-1-cyclohexanol
d) 1-hydroxyl-3-methylbenzene

18. How does the solubility of carboxylic acids in water change as the hydrocarbon chain length increases?

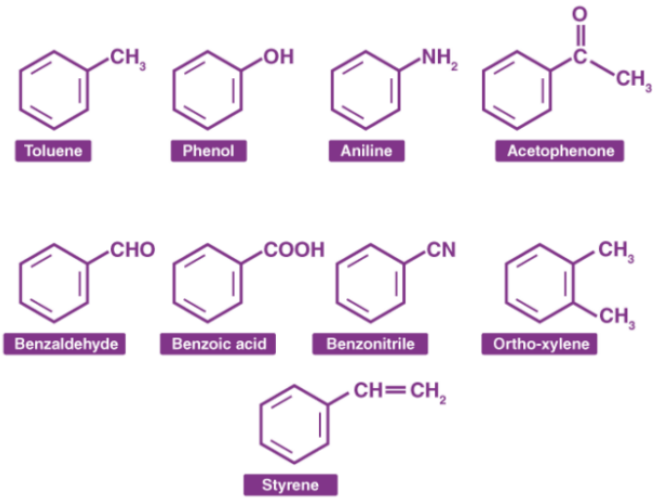
a) Increases
b) Decreases
c) Remains the same
d) Fluctuates irregularly

19. Which of the following is a tertiary amine?

a) $CH_3CH_2NH_2$
b) $(CH_3)_2NH$
c) $(CH_3)_3N$
d) NH_3

20. Which structure represents an amide?

a) R–CO–NH_2
b) R–COOH
c) R–O–R
d) R–NH_2



Substituent (Group)	Parent name when principal	Priority
–COOH	benzoic acid	**Highest**
–OH	phenol	High
–NH_2	aniline	Medium
–CH_3	toluene	Low
–Br, –Cl	halo- (e.g., bromobenzene, chlorobenzene)	Lowest

21. Draw line-angle formulas for the following compounds:

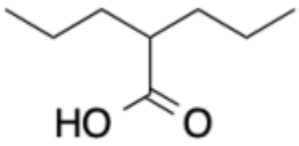
a) 2-Methylcyclopentanethiol

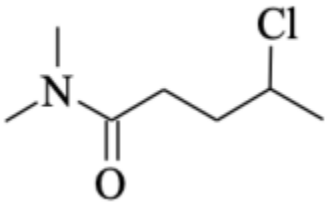
b) 3,3-Dibromo-2-pentanone

Name

	A	B	C	D	E		A	B	C	D	E
1	○	○	○	○	○	11	○	○	○	○	○
2	○	○	○	○	○	12	○	○	○	○	○
3	○	○	○	○	○	13	○	○	○	○	○
4	○	○	○	○	○	14	○	○	○	○	○
5	○	○	○	○	○	15	○	○	○	○	○
6	○	○	○	○	○	16	○	○	○	○	○
7	○	○	○	○	○	17	○	○	○	○	○
8	○	○	○	○	○	18	○	○	○	○	○
9	○	○	○	○	○	19	○	○	○	○	○
10	○	○	○	○	○	20	○	○	○	○	○

22. Provide the IUPAC name for the following compounds:





_____ _____

24. Explain why tertiary alcohols do not undergo oxidation.